

- experiencing a rash on your skin. What is the probability that you have dengue fever? Is the information provided sufficient to answer the question? If not, explain why.
6. You are given a set of documents and asked to cluster them based on document level similarity. You are familiar with different clustering algorithms such as 'kmeans', so you are confident that you can do the clustering. However, the interviewer now adds a second part to the problem, in which he asks you to label the clusters. How will you generate labelled clusters? What are the means of generating meaningful labels for the clusters? Can the clusters be labelled post the clustering process, prior to it, or during the process? Describe your solution and justify why you made that particular choice.
 7. With elections coming around in a few of the Indian states, poll predictions have become quite a hot topic. However, there has only been limited success in poll prediction. For instance, the past few years have been quite unsuccessful for many of the data scientists who are into poll prediction. Neither Brexit nor Trump's victory were anticipated by most of those making the predictions. Given this context, you are asked to predict the party which will win the majority in each of the five states going to the polls in the coming months. For any prediction, you need to build models that can analyse the data and predict. What are the various data inputs that you will be interested in getting? Or, in simplistic terms, what are some of the obvious features that you would like to extract from the data to be able to predict the winning party in a state election? Is this a classification problem or a clustering one? Why do you think poll prediction is a hard problem?
 8. Given a classifier, can you define the terms 'sensitivity' and 'specificity' of your system? Let us assume that you have built a classifier that can predict whether a tumour is benign or malignant based on certain characteristics of such growths. Your intention is to build a classifier that can detect all true malignant tumours, even if some of the benign tumours get falsely reported as malignant. Which of the two metrics, 'sensitivity' or 'specificity', should you try to improve in this case? Now assume that you have been asked to use a classifier to predict whether a student will pass the examination or not. Your system should be designed such that it does not report a failing student as having passed, even if it wrongly predicts certain passing students as potential failures. In that case, which of the two metrics, 'sensitivity' or 'specificity', should you try to improve?
 9. You are given a large text document and asked to create a summary of it. There are two types of summarisation algorithms—extractive and abstractive. In extractive summarisation, the summary will contain sentences extracted from the text. In abstractive summarisation, the summary will contain the gist of the original text, without including the exact sentences from the text. What kind of summarisation algorithm would you choose? Explain the rationale behind your choice. Now assume there is a real-time conversation that is occurring. How would you create a running summary of it?
 10. Let us consider a binary classification problem in which you are asked to label emails as either spam or non-spam. You are given a large enough training data set (a 100,000 emails) in which the emails are already classified as spam or non-spam. Using this training data, you are asked to train a binary classifier that can mark an incoming email as spam or non-spam. However, there is one problem with the training data that you had been given. Among the 100,000 training data emails, only 100 are labelled as spam. All other training data are instances in the non-spam category. What would be the problem if you train your model with such an unbalanced training data set? How will you overcome this issue?
 11. In solving classification and regression problems in machine learning, great emphasis is laid in feature engineering, whereby the right set of features is carefully selected among all the features of the given data samples. For instance, consider a linear regression problem in which each sample of input data has four features: X_1 , X_2 , X_3 and X_4 . X_3 and X_4 are linearly dependent on X_2 . X_1 is linearly independent of all the other features. For example, assume that the regression problem is to predict house prices. X_1 is the type of locality, X_2 is the size of the house, while X_3 and X_4 are the size of the bedroom and size of the external yard, respectively. You have created two classifiers, G1 and G2, such that G1 uses all four features and G2 uses only X_1 and X_2 . Will G2 be more accurate than G1? What are the problems with G1 in comparison with G2?
 12. Can you explain the terms 'domain adaptation' and 'transfer learning'? What is the difference between the two?
 13. Why are LSTMs preferred as convolutional neural networks in text classification systems?
 14. How is the attention mechanism used in encoder-decoder neural networks for sequence-to-sequence generation? Can you explain what is meant by intra-attention and what is meant by inter-attention in sequence-to-sequence models?
 15. How does one avoid over-fitting in deep neural networks? What are the different methods used in practice?
- If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Wishing all our readers happy coding until next month and a very happy new year ahead! **END** 🐧

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